

Ejercicio 6

Dadas las siguientes funciones:

```
zip :: [a] -> [b] -> [(a,b)]
{Z0} zip = foldr (\x rec ys ->
  if null ys
  then []
  else (x, head ys) : rec (tail ys))
  (const [])
```

```
zip' :: [a] -> [b] -> [(a,b)]
{Z'0} zip' [] ys = []
{Z'1} zip' (x:xs) ys = if null ys then [] else (x, head ys):zip' xs (tail ys)
```

Demostrar que $zip = zip'$ utilizando inducción estructural y el principio de extensionalidad.

Busco probar que $zip = zip'$

Por inducción estructural en listas

Por extensionalidad alcanzo probar que

$\forall xs :: [a], \forall ys :: [b]. zip\ xs\ ys = zip'\ xs\ ys$

Hago inducción sobre la lista xs

$P(xs) : \forall ys :: [b]. zip\ xs\ ys = zip'\ xs\ ys$

Caso base $xs = []$

$P([]) : zip\ []\ ys = zip'\ []\ ys$

$zip\ []\ ys$
 $=_{\{Z0\}} foldr\ (\lambda x\ rec\ ys \rightarrow$
 if null ys
 then []
 else (x, head ys) : rec (tail ys))
 (const []) [] ys

$=_{\{FRO\}} (const [])\ ys$

$=_{\{const\}} []$

$zip' [] ys$

$= []$

Wegso vale el caso base

Paso Inductivo

Busco probar que $\forall xs :: [a], \forall x :: a. P(xs) \Rightarrow P(x:xs)$

Defino mi $\{HI\}$ $zip\ xs\ ys = zip'\ xs\ ys$

Busco probar $P(x:xs): zip\ (x:xs)\ ys = zip'\ (x:xs)\ ys$

$zip\ (x:xs)\ ys$

$= foldr (\lambda x\ rec\ ys \rightarrow$
 $\quad \text{if null } ys$
 $\quad \text{then } []$
 $\quad \text{else } (x, head\ ys) : rec\ (tail\ ys))$
 $\quad (const\ [])\ (x:xs)\ ys$

$= foldr\ f\ (const\ [])\ xs)\ ys$

$=$ if null ys

then []

else (x, head ys) : (foldr f (const []) xs) (tail ys)

$=$ if null ys

then []

else (x, head ys) : zip' xs (tail ys)

$$\{z'\} \stackrel{=}{=} \text{zip}'(x:xs) \text{ y}$$

luego queda probado el caso inductivo

$$\circ \text{zip} = \text{zip}'$$